

AISHWARYA NAYAK

Senior AI Engineer | Applied AI | Backend Systems

Bengaluru, India | +91 8989438322 | ashnayakwork@gmail.com | LinkedIn: <https://www.linkedin.com/in/aishwarya-nayak-/>
| GitHub: <https://github.com/A1SHWARYANAYAK> | Portfolio: <https://a1shwaryanayak.github.io/>

SUMMARY

Senior Software Engineer and Applied AI Engineer with 5 years of experience delivering distributed systems, cloud-native services, and AI-driven applications across USA and India teams. Backed by an MS in Data Science from the University at Buffalo (SUNY), with expertise spanning backend engineering, applied AI, frontier-model evaluation, and software-engineering benchmarks. Experienced in building scalable APIs, reproducible testing frameworks, and reliable AI systems while combining strong software engineering fundamentals with modern machine learning practices.

TOOLS & TECHNOLOGIES

Programming Languages: Python, Java, Scala, JavaScript, TypeScript, SQL, Bash

Artificial Intelligence & Machine Learning: Large Language Models (LLMs), Agentic AI, AI Evaluation, AI Benchmarking, Prompt Engineering, Multimodal AI, Retrieval-Augmented Generation (RAG), AI Safety, Human-in-the-Loop (HITL), Scientific Integrity Evaluation, Machine Learning, Model Evaluation, Error Analysis

Frameworks & Libraries: LangChain, LangGraph, FastAPI, Pydantic, OpenAI API, Gemini API, Vertex AI

Software Engineering: Backend Development, REST APIs, Microservices, Distributed Systems, Event-Driven Architecture, Software Architecture, Open-Source Development, Feature Development, Bug Fixing, Runtime Debugging

Testing & Benchmark Engineering: Software Engineering Benchmarks, Autonomous Coding Agent Evaluation, Behavioral Testing, Regression Testing, Fail-to-Pass (F2P) Testing, Pass-to-Pass (P2P) Testing, Pytest, Deterministic Evaluation Pipelines, Oracle Implementation Development

Cloud & DevOps: Docker, Kubernetes, Git, GitHub, GitHub Actions, GitHub CLI, Jenkins, AWS, Google Cloud Platform (GCP), Linux

Databases & Search: MySQL, Apache Solr, ChromaDB, Sentence Transformers

Developer Tools: Harbor, n8n, MCP, Slack, JSON, TOML, LaTeX, KaTeX

PROFESSIONAL EXPERIENCE

Handshake AI — AI Evaluation Engineer

Remote (USA & India) | September 2025 – Present

Worked across multiple AI research initiatives focused on frontier large language models, multimodal systems, AI safety, software engineering benchmarks, and autonomous coding agent evaluation.

Software Engineering Benchmark Development

- Designed and authored realistic software engineering benchmark tasks using production-scale open-source repositories to evaluate autonomous AI coding agents.
- Identified real-world bug fixes, feature enhancements, API migrations, and repository-specific engineering challenges suitable for benchmark creation.
- Developed implementation-agnostic benchmark specifications with clearly defined behavioral requirements and evaluation criteria.
- Built deterministic Fail-to-Pass (F2P) and Pass-to-Pass (P2P) test suites supporting multiple valid implementations while preventing solution leakage.
- Implemented production-quality reference solutions, benchmark metadata, debugging hints, evaluation artifacts, and reproducible task packaging.
- Engineered reproducible Docker environments and Harbor-based evaluation pipelines to ensure deterministic execution across diverse repositories.
- Validated benchmark quality against frontier coding models by analyzing agent failure modes, identifying shortcut solutions, and iteratively improving benchmark robustness.

AI Evaluation & Code Quality Assessment

- Evaluated AI-generated software across Python, JavaScript, TypeScript, R, C++, and additional programming languages using execution-based evaluation frameworks.
- Executed generated applications within isolated Docker environments to validate runtime behavior, algorithmic correctness, software design, explanation quality, and instruction adherence.
- Identified runtime failures, logical inconsistencies, hallucinations, incomplete implementations, hidden execution errors, and behavioral regressions.
- Produced reproducible evaluation evidence using execution logs, terminal outputs, structured rubrics, and mirrored evaluation platforms while adapting to evolving project standards.

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Prompt Engineering & Reasoning Evaluation

- Designed challenging multimodal and text-only reasoning tasks to evaluate frontier large language models across computer science and related technical domains.
- Authored image-dependent and text-based prompts following structured evaluation methodologies and standardized quality frameworks.
- Produced complete human-authored reference solutions, structured answer schemas, reproducibility metadata, and detailed technical explanations.
- Validated prompts across multiple frontier models to identify genuine reasoning failures while eliminating ambiguity and retrieval-based errors.
- Reviewed prompts using structured quality rubrics to identify plagiarism, ambiguity, policy violations, invalid evaluation methodology, and overall benchmark quality.

AI Safety & Scientific Integrity Evaluation

- Audited autonomous AI agent trajectories by evaluating problem formulation, planning, implementation, experimentation, and overall methodological integrity.
- Detected scientific integrity violations including data leakage, metric hacking, evaluation contamination, plan-code divergence, constraint violations, tool misuse, and evidence tampering.
- Compared natural-language reasoning with executed code to verify behavioral consistency throughout multi-step experimental workflows.
- Produced evidence-backed integrity assessments and realistic adversarial examples to strengthen AI safety evaluation datasets.

Machine Learning Evaluation

- Evaluated competing machine learning implementation strategies across feasibility, implementation complexity, novelty, and expected impact.
- Analyzed objectives, datasets, evaluation metrics, and implementation approaches to determine technically appropriate algorithmic solutions.
- Identified invalid evaluation tasks involving duplicate algorithms, non-machine-learning_objectives, and policy violations.

Multimodal AI & Data Curation

- Contributed to multimodal dataset generation initiatives through standardized collection, quality assurance, and compliance workflows.
- Captured high-quality multimodal datasets following strict privacy, safety, consent, and geographic requirements.
- Performed quality validation to maximize downstream usability for computer vision, navigation, and multimodal AI model training.

Reality AI Lab — Generative AI Engineer

Buffalo, New York, USA | Mar 2025 – Sep 2025

- Developed a retrieval-augmented AI teaching assistant enabling interactive academic dataset querying using Python, LangChain, and OpenAI APIs.
- Built scalable ingestion and preprocessing pipelines supporting long-context LLM applications, reducing input errors by 15%.
- Deployed models using GCP Vertex AI and containerized services, automating model training and production inference workflows.
- Integrated logging, evaluation, and monitoring frameworks to improve reliability and accelerate root-cause analysis.
- Collaborated with backend engineers to design modular APIs supporting knowledge graph updates and prompt orchestration workflows.
- Contributed to the end-to-end lifecycle of AI applications spanning data ingestion, model deployment, inference, and observability.

Mocapay — Software Engineer, Backend

Delhi, India | Aug 2021 – Jun 2023

- Developed and maintained reactive Scala microservices supporting fintech products generating over \$10M in annual revenue.
- Designed and implemented RESTful APIs for large-scale transaction processing systems, improving reliability and customer experience.

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- Integrated Apache Kafka for asynchronous communication and event-driven workflows, enhancing real-time transaction capabilities and system resilience.
- Streamlined CI/CD pipelines using Jenkins and Docker, reducing deployment time by 30%.
- Enhanced platform robustness and security in accordance with Vanta recommendations, contributing to SOC 2 compliance.
- Improved search capabilities by integrating Apache Solr, increasing search efficiency by 25%.
- Led code reviews and promoted software engineering best practices to improve maintainability and code quality.

Tata Consultancy Services (TCS) — Software Engineer, Backend

Indore, India | Jan 2020 – Jul 2021

- Led development of a finance platform supporting management of over 1,000 contracts and improving operational efficiency.
- Designed backend services and RESTful APIs for contract creation, validation, and payment calculations.
- Managed cloud deployments on AWS and GCP, reducing infrastructure costs by 15%.
- Optimized SQL queries and improved database performance by 20%.
- Implemented security controls and role-based access mechanisms using Fortify and SonarQube, reducing vulnerabilities by 40%.
- Automated testing and deployment pipelines using Jenkins, significantly shortening release cycles.
- Collaborated with cross-functional teams during sprint planning, backlog refinement, and product roadmap discussions.

SELECTED PROJECTS

Enterprise Knowledge Copilot | RAG Knowledge Assistant [GitHub]

Tech Stack: Python, LangChain, Vector Databases, OpenAI APIs, FastAPI

- Built an enterprise-grade Retrieval-Augmented Generation (RAG) platform for contextual question answering and knowledge discovery across organizational documents.
- Designed scalable ingestion and retrieval pipelines supporting structured and unstructured data sources.
- Implemented semantic search and source-grounded response generation to improve answer quality and reduce hallucinations.
- Developed modular APIs and extensible workflows supporting future tool integrations and agent-based interactions.
- Enabled efficient knowledge access and productivity improvements through context-aware conversational interfaces.

CodeSage | Multi-Agent Repository Intelligence Platform [GitHub]

Tech Stack: Python, LangGraph, Pydantic, GitHub APIs

- Developed a LangGraph-based multi-agent system for automated repository analysis and architecture understanding.
- Built specialized agents for dependency inspection, quality assessment, security analysis, and documentation generation.
- Implemented shared-state orchestration and structured outputs using Pydantic to improve reliability and maintainability.
- Automated report generation workflows to accelerate onboarding and analysis of unfamiliar codebases.
- Designed extensible agent workflows for evaluating repositories and enhancing developer productivity.

Transcript Intelligence | AI-Powered Transcript Analytics Platform [GitHub]

Tech Stack: Python, Jupyter Notebook, Pandas, OpenRouter, Claude Sonnet 4.5, structured JSON outputs, Matplotlib

- Built an AI-powered transcript analytics pipeline that classifies business conversations, analyzes sentiment, and extracts cross-functional insights from unstructured call data.
- Designed a hybrid topic classification workflow combining LLM-generated taxonomy discovery with human review to improve consistency and business relevance.
- Implemented multi-dimensional sentiment analysis covering customer sentiment, urgency, resolution status, and sentiment trajectory rather than relying on a single polarity label.
- Developed a cross-functional incident timeline that linked support, external customer, and internal engineering conversations to reconstruct outage impact across teams.
- Added reproducible outputs, validation artifacts, and notebook-based reasoning to support transparent analysis and portfolio presentation.

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FlowForge AI | AI Workflow Automation Platform [\[GitHub\]](#)

Tech Stack: n8n, Python, APIs, LLM Integrations

- Developed an AI-driven workflow automation platform enabling orchestration of multi-step tasks through modular and event-driven pipelines.
- Built reusable workflows integrating external APIs and large language models to automate repetitive processes and improve productivity.
- Designed extensible automation patterns supporting tool interactions, conditional execution, and multi-stage task coordination.
- Leveraged low-code orchestration principles to simplify integrations and accelerate workflow development.
- Enhanced operational efficiency through reusable components and scalable workflow design.

Log-Report | AI-Powered Log Analysis and Incident Reporting System [\[GitHub\]](#)

Tech Stack: Python, OpenAI API, LangChain, Pydantic, JSON, Logging, Regex, Async Processing

- Built an AI-powered log analysis pipeline that transforms raw logs into structured incident reports.
- Implemented intelligent parsing, error clustering, root-cause extraction, severity classification, and timeline construction.
- Generated Markdown and JSON reports for production incident review, debugging, and support automation.

EDUCATION

Master of Science (MS) in Data Science

University at Buffalo (State University of New York, SUNY Buffalo)

Aug 2023 - Feb 2025
New York, US

Bachelor of Technology (B.Tech) in Electronics and Telecom Engineering

Symbiosis International University

Jul 2015 - Apr 2019
Pune, IN